



Genome-wide analysis of DUF221 domain-containing gene family in common bean and identification of its role on abiotic and phytohormone stress response

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Received: 8 March 2022 / Accepted: 30 May 2022 / Published online: 2 July 2022
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Abstract The domain of unknown function (DUF221 domain-containing) proteins regulates various aspects of plant growth, development, responses to abiotic stresses, and hormone transduction pathways. A comprehensive genome-wide analysis was performed in its genome to understand the role of *DDP* genes (DUF221) in the common bean (*Phaseolus vulgaris* L.). A total of 12 *DDP* genes were identified and distributed in 8 chromosomes in the common bean genome. The physical and biochemical characteristics of amino acids, motif and intron–exon structure, and cis-regulatory elements of *DDP* members were determined. Phylogenetically all *PvDDPs* were

clustered into nine clades, subsequently supported by their gene structure and conserved motifs distribution. The *PvDDPs* contained various cis-acting elements involved in plant responses to abiotic and various phytohormones stresses. A total of 45 different cis-regulatory elements in the putative promoter regions of the *PvDDPs* were identified. ERE and ABRE were discovered to be present in all *PvDDPs*, indicating that they may be regulated by ethylene and ABA, both of which are strongly associated with biotic stress response in plant species. Additionally, *PvDDPs* were targeted by multiple miRNA gene families as well. In this context, the most targeted *DDP* family members are *PvDD10* and *PvDDP11*. The miRNA target analysis showed that Pvu-miR2594, Pvu-miR169, Pvu-miR2584, Pvu-miR530, Pvu-miR156, and Pvu-miR2592 target these genes. There is a strong correlation between abiotic stress and *PvDDPs* expression in both leaf and root tissues. *PvDDP11* is the unique and highest upregulated gene with hormone treatment and abiotic stress among all the members. Expression of the *PvDDP11* gene indicated a strong correlation with drought and salt stress in the common bean roots and leaves, respectively. In **conclusion**, this study predicted that the putative *DDP* genes might help improve abiotic and phytohormone tolerance in common bean.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10722-022-01421-7>.

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Keywords ABA · Abiotic stress · DUF221 ·
miRNA · IAA · RSN1_7TM

Introduction

Common bean (*Phaseolus vulgaris* L.) is considered to be one of the most important and affordable crops that meet a significant part of the protein, carbohydrate, vitamin, and micro-mineral needs in human nutrition (Broughton et al. 2003). Its nitrogen-fixing ability and short growth cycle make the common bean an excellent crop species for bioproductivity enhancement and marginal land reclamation throughout Asia, Eastern Africa, and Latin America (Cortes 2013). Turkey is an important producer of annual fresh common bean in the world, and almost 60% (360 Mt) of its production (Hiz et al. 2014) is dominated in the Black Sea region, where the soil salinity levels can reach up to 2–4 dS/m (Niron et al. 2020). However, it is known that the common bean is highly sensitive to salinity, and even one dS/m salinity level could cause a 20% reduction in its productivity (Bayuelo Jimenez et al. 2012). It has been shown that the salinity problems can be overcome by developing and cultivating new salt-tolerant common bean genotypes like Ispir. This common bean cultivar has been reported to block Na^+ accumulation significantly in its leaves and roots compared to susceptible genotypes under salinity stress (Niron et al. 2020). Despite the existence of such salt-tolerant superior genotypes, salinity in agricultural areas is estimated to increase continuously with the rise in drought, excessive evaporation, and inadequate precipitation, all of which started to happen with global warming-dependent climate change (Jamil et al. 2011). It was estimated that the accumulation of salts and ions in upper soil layers combine the osmotic stress and ion toxicity in plants with further global warming in the future (Kavas et al. 2021). Tolerant genotypes like Ispir were found to cope with these abiotic stresses by physiochemical alterations (Niron et al. 2020) and molecular regulations (Hiz et al. 2014). Therefore, further molecular and genome-based studies on these salt-tolerant common bean genotypes should be carried out to identify their main genetic regulations and pathways behind the tolerance. This knowledge could then be used to expand their tolerance capacity and develop new plant cultivars resistant to abiotic stress tolerance.

Domains of unknown function (DUFs) constitute the largest set of families in protein databases (Pfam) and are generally utilized to describe any proteins of unknown function (Bateman et al.

2010). In recent years, the number of DUF families has increased dramatically with the number of sequenced species genomes. Almost 22% of the Pfam database (version 32.0) was constituted by DUF families recently (Yang et al. 2020). Many genomic analyses and reports were prepared to investigate the origin, evolution, and biological functions of the DUF families in many plants such as *Oryza sativa*, *Arabidopsis thaliana*, and *Zea mays* (Cao et al. 2009; Hou et al. 2014; Li et al. 2016, 2017, 2018; Li Lihua et al. 2018; Nabi et al. 2021; Nietzsche et al. 2014; Vaattovaara et al. 2019; Wang et al. 2014). In these studies, DUF genes were found to have important regulatory functions in plants during phytohormone and abiotic stress responses, especially drought and salinity. The DUF221 domain-containing proteins (DDPs) could be a good candidate for searching in the common bean genome due to their osmosensitive ion-permeable channel roles in plants (Hou et al. 2014). This domain has been identified in stress-responsive genes of several plants, such as the dehydration-responsive gene (ERD4) of *Brassica juncea* (Rai et al. 2012) and drought-regulated genes of rice (Shaik RafiRamakrishna Wusirika 2012). DUF221 domains were previously associated with the osmotic shock-responsive influx cation channel gene in Arabidopsis, and it was found to enable the permeability of Ca^{2+} ions under hyperosmotic shock (Hou et al. 2014). Similarly, Yuan et al. (2015) have characterized a DUF221 domain-containing hyper-osmolality-gated calcium-permeable channel protein in rice and revealed its functional role in regulating intracellular Ca^{2+} levels under hyperosmotic stress. Ganie et al. (2017) analyzed the DDPs family in rice species and identified many of its salinity-responsive members. DDPs in tomato contained various cis-acting elements involved in plant responses to abiotic and various phytohormones stresses (Waseem et al. 2021). As can be understood from this literature, DDP genes possess an important regulatory and functional role in plants, especially under hyperosmotic stress. However, the existence and function of this important domain remain to be determined in common bean. Therefore, the major aim of this current study is to perform a comprehensive study of DDP gene family in the salt-tolerant common bean genotype, Ispir. These genes were firstly predicted by in-silico

analysis in common bean genome and then confirmed with expression analysis in different organs/tissues. Moreover, Ispir genotype was treated with salinity, drought (PEG), and phytohormone stresses (ABA and IAA) and qPCR assays were applied to the stress treat tissues to understand the alteration of *DDP* genes under these stresses and to evaluate their role in stress tolerance. Thus, the study provided the first data on the functional validation of the common bean DUF221 domain-containing genes and their role in tolerance under stressful conditions.

Materials and methods

Identification of *DDP* gene family members in common bean genome

In the current study, complementary methods were performed to identify the DUF221 domain-containing protein (DDPs) family in the common bean genome. Firstly, Hidden Markov Model (HMM) profile was used to obtain DDPs (PF03181) from Pfam30.0 (Finn et al. 2016). Then this profile was used as a query to discard and form a candidate DDPs list from the common bean genome using HMMER3.0 (Finn et al. 2015). In addition to this first step, a keyword search was carried out in the Phytozome V13 database (<https://phytozome.jgi.doe.gov/pz/portal.html>) to check if there are overlooked DDPs (Kavas et al. 2017). Pfam (<http://pfam.xfam.org/>) and SMART databases (<http://smart.embl-heidelberg.de>) were further used to confirm DDPs' existence in the common bean genome. Gene sequences that do not match with the DUF221 domain or ones including an uncertain domain were removed from the list of DDPs. In the final step, chromosomal locations of the candidate genes and their protein-coding sequences were obtained from the Phytozome v13 database (<https://phytozome-next.jgi.doe.gov/>). Molecular and physicochemical parameters such as molecular weight (Mw), isoelectric point (pI), instability index, and hydropathy scale (GRAVY) were predicted by using the ProtParam tool located on the ExPASy website (<https://web.expasy.org/protparam>). Moreover, the solubility of the candidate *PvDDPs* was estimated by the SoluProt tool according to their protein-coding

sequence (Hon et al. 2021). These analyses were enabled to create a final candidate list of common bean' DDPs called *PvDDPs* from now on.

Sequence analysis, gene duplication, and phylogeny of *PvDDP* genes in common bean

Gene sequences of the candidate *PvDDPs* and their coding parts were downloaded from the Phytozome v13 database (<https://phytozome-next.jgi.doe.gov/>) to find out their intron–exon structures. Conserved motifs in *PvDDPs* were analyzed via the MEME suite tool by choosing 15 repeated motifs. CLC Genomics Workbench tool was utilized to estimate the positions of conserved DUF221 domains and the signal peptides on these proteins. The intron/exon structure and conserved motifs of *PvDDPs* were represented using Gene Structure View of TBtools software (<https://github.com/CJ-Chen/TBtools>) (Chen et al. 2018). The locations of *PvDDPs* on the common bean genome were determined with Gene on Genome from Fasta application using Blast. Their distribution on the common bean chromosomes was visualized by using TBtools software (<https://github.com/CJ-Chen/TBtools>). Approximately 2000 bp upstream region of each *PvDDPs* was analyzed with PlantCARE Database (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) to find out the promoter region and regulatory sequences (TATA-box and Cis elements).

In addition to *PvDDP*' identification in the common bean genome, miRNAs regulating the expression of these genes were also investigated in the current study. Therefore, candidate miRNAs were firstly retrieved in the common bean genome via Plant miRNA Encyclopedia (PmiREN, <http://www.pmiREN.com/>) (Guo et al. 2020). Putative miRNAs targeting *PvDDPs* were then predicted via the psRNATarget tool depending on miRNA-target complementary score patterns (<http://plantgrn.noble.org/psRNATarget>) (Dai et al. 2018) and confirmed with Miranda tool based on default parameters (Enright et al. 2003).

Gene duplications of the *PvDDPs* were also predicted in the current study by using Plant Duplicate Gene Database (PlantDGD) (Qiao et al. 2019). The duplicate genes located on the same chromosome were accepted as tandem repeats, and another BLASTP search was further utilized to identify segmental duplications of *PvDDPs* in the common bean. The latter analysis was dependent on five matches

between two duplicates, and according to *E*-value ($\leq 1e-05$), the MCScan tool was utilized to determine their collinear blocks (Wang et al. 2012).

To estimate the evolution of the *PvDDP* genes, a ratio between nonsynonymous mutation rate and synonymous mutation rate (Ka/Ks) was calculated via TBtools software. *PvDDP* proteins were blasted against the Phytozome v13 database, and the results over the 60% sequence similarity threshold were used for this evolutionary estimation. This similarity data was then used to create a tab-delimited text file required to calculate Ka/Ks in TBtools software. In addition to evolutionary relation, the phylogeny of the *PvDDPs* was also investigated within the study. The multiple sequence alignment of common bean DUF221 domain-containing proteins was firstly performed using Muscle. Then phylogenetic trees were constructed via MEGAX software by using the neighbor-joining (NJ) method with the bootstrap test (1,000 replicates) and Jones-Taylor-Thornton (JTT) model. These phylogenetic dendrograms were visualized via ITOL v3 (<http://itol.embl.de/>). All these evolutionary, multiple alignments and phylogenetic analyses of *PvDDPs* were predicted by considering amino acid sequences of DUF221 from various species such as *Glycine max*, *Arabidopsis thaliana*, *Cucumis sativus*, *Medicago truncatula*, *Zea mays*, and *Oryza sativa*. Lastly, the Dual Synteny Plotter software (<https://github.com/CJ-Chen/TBtools>) Liu et al. (2017) was used to represent the synteny relationship of the orthologous DUF221 genes between common bean and *Arabidopsis* as well as common bean and rice.

Prediction of protein structure, protein–protein interaction and cellular localization of *PvDDPs* in common bean

Plant-mPLoc server (<http://www.csbio.sjtu.edu.cn/bioinf/plant-multi/>) was employed to predict subcellular localization (ChouShen 2008). Proteins sequences of *PvDDPs* were retrieved from the Phytozome database. To identify the 3D structure, *PvDDPs* were investigated in the Phyre2 server (Protein Homology / Analogy Recognition Engine) with an "intensive" mode (Kelley et al. 2015). Trans-membrane helical domains were predicted using TMHMM (<https://services.healthtech.dtu.dk/service.php?TMHMM-2.0/>) (Krogh et al. 2001). In order to explore the potential

interaction network associated with *PvDDPs* with the other proteins, a protein–protein interaction (PPI) network was constructed by The Search Tool for the Retrieval of Interacting Genes (STRING) database (<https://string-db.org/cgi/input.pl;version:11.5>) employing the involved mechanism with Pfam, UniProt, and InterPro database, GO biological process analyses, and KEGG pathway. *PvDDPs* were named by our inputs in the PPI map, and others were introduced based on names or locus numbers.

Plant material and stress treatments

The common bean genotype Ispir was selected in stress treatments due to its wide cultivation in Turkey and its confirmed resistance to salinity. The seed of Ispir was sterilized with 5% sodium hypochlorite before being sown in pots filled with sterile vermiculite. After germination, the plants were grown in a fully controlled growth chamber at 24 °C under a 16/8 photoperiod. Four weeks after planting, Ispir plants were subjected to drought, salinity, and hormone (ABA and IBA) dependent stress. The drought and salinity stress were created on the plants by adding polyethylene glycol 6000 (20%) and 200 mM NaCl into the Hoagland solution used for irrigation, respectively. 100 µM ABA and 100 µM IAA were applied to the plants by spraying them to the leaves. The plants were fully harvested from the pots after the completion of stress treatments. Stress and hormone-treated leaf and root samples were collected at 6, 12, 24, 48, 60th hour and seventh day after stress treatment and stored at -80° C until their utilization for RNA isolation. Plants grown at 24 °C under a 16/8 photoperiod without any stress application were used as a control throughout the study.

RNA isolation and gene expression analysis

RNA isolations on the stress and hormone-treated leaves and roots of Ispir were carried out by RNeasy Plant Mini Kit (Qiagen) and its manufacturer protocol. The quality, integrity, and quantity of the isolated total RNAs were evaluated by NanoDrop™ 2000 / 2000c spectrophotometer and agarose gel electrophoresis. The iScript™ cDNA Synthesis Kit was used to synthesize the cDNA from the isolated total RNAs. Expression analysis of *PvDDP* genes in the stress and hormone-treated leaves and roots of Ispir

genotype was carried out on the Agilent Mx3000P with Solis BioDyne 5×HOT FIREPol® EvaGreen® qPCR Mix Plus (ROX). The conditions of the qRT-PCR experiments were as follows; denaturation at 95°C for 15 min; 35 cycles of 95°C for 15 s, 60°C for 20 s, and 72°C for 20 s. The common bean 18S rRNA gene was used as an internal control (Kavas et al. 2016). Relative expression of all *PvDDP* genes (*PvDDP1*–*PvDDP12*) was calculated according to $2^{-\Delta\Delta C_t}$ method. The primer sequences utilized in the qRT-PCR experiment was represented in Table S1.

In silico expression analysis

The expression profiles of *PvDDP* genes were evaluated employing the RNA-seq data sets, consisting of the transcriptional response to drought and salt stress. To calculate the *DDP* genes' responses against different conditions in various tissues, NCBI Sequence Read Archive (SRA) was used to download the raw RNA-seq data-sets of accession numbers named PRJNA508605 and PRJNA327176 (drought), PRJNA574280, PRJNA691982, PRJNA656794, PRJNA558376 (salt), and PRJNA574280 (pathogen). The downloaded files were processed with the help of Galaxy Europe Server (usegalaxy.eu) and CyVerse (<https://www.cyverse.org/>), including virtual bioinformatics tools. The mapping of reads was established via the HISAT2 tool. Transcript assembly and differential expression analyses were performed in Stringtie 1.3.3 and Ballgown, respectively. A gene was accepted as differentially expressed (DEG) if the p -value < 0.05 and a $\log_2 FC > 1$. Expression analysis was summarized by preparing a $\log_2 FC$ based Heatmap via TBTools. A total of eight transcriptome comparisons were used in this heat map. The first one was the transcriptome comparison made between salt-treated leaf explants of T43 (salt-sensitive genotype) and Ispir (salt-tolerant genotype) (PRJNA656794). The second comparison included the root samples for the same genotypes under the same salt stress condition. Similarly, third transcriptome data comparing only the salt-treated and control roots of Ispir genotype was used to evaluate the expression level of *PvDDPs*. The fourth comparison was carried out on the transcriptome data (PRJNA558376) obtained from buds of Ispir and T43 genotypes subjected to salt stress. The fifth transcriptome comparison included the salt-treated sprout of Ispir genotype

(PRJNA691982). The sixth data contained the transcriptome comparison between drought-treated Pinto Saltillo (drought-tolerant) and its control counterpart on the whole common bean plant (PRJNA508605). The seventh experiment comprised RNA sequencing (RNA-Seq) on drought-treated roots and leaves of two contrasting drought-responsive genotypes; drought-tolerant BAT 477 and drought-sensitive Pérola. Regulation of *PvDDPs* expression under biotic stress conditions was also evaluated in the current study with transcriptome data (PRJNA574280) obtained from a fungal pathogen (*Sclerotinia sclerotiorum*) treated common bean genotypes. Phytozome v13 was used to identify tissue-specific expression patterns of *PvDDPs*.

Results

Genome-wide identification of *PvDDP* gene family in common bean

In the current study, *PvDDP* genes having the DUF221 domain were investigated throughout the common bean genome. Their expression levels were examined to find out their potential role in the stress response. Various web-based software, including the HMM search tool, was utilized to identify *PvDDP* family members in the common bean genome. This further search on the Pfam and SMART confirmed the *PvDDP* gene family in common bean and excluded the genes that do not have the DUF221 domain. All these analyses revealed 12 common bean *PvDDP* gene family members named *PvDDP1*–*PvDDP12* according to their locations on chromosomes, as shown in Fig. 1. The locations of the *PvDDPs* in the common bean genome dispersed into eight chromosomes out of the 11. Among all *PvDDPs*, eight were located on the upper parts of chromosomes, and three were found in the below parts, while just one was found in the middle position.

The name of the genes and their chromosome location were provided in Table 1 with the genomic and protein properties. The gene and protein structures of *PvDDP* family members varied significantly within each other. When the length of the genes was compared, it was determined that there was an 831 bp difference between the shortest (2307 bp-*PvDDP7*) and the longest (3138 bp-*PvDDP5*) members of the bean.

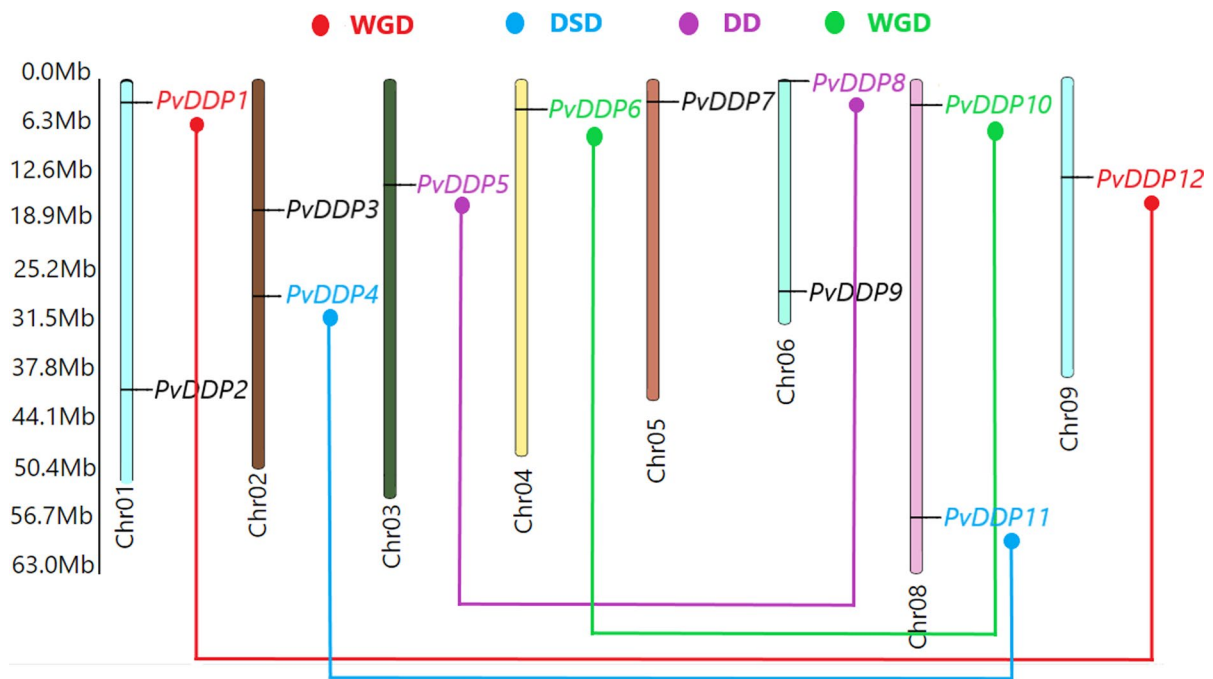


Fig. 1 Chromosomal location of 12 common bean RSN1_7TM domain-containing *PvDDP* genes. Based on their physical location, the genes are mapped over the eight linkage groups: Chr01, Chr02, Chr03, Chr04, Chr05, Chr06, Chr08, and Chr09. The chromosome number is represented at

the bottom of each chromosome. The identity of each linkage group is provided at the top of each bar: WGD, whole-genome duplication; DSD, dispersed segmental duplication; DD, DNA-transposed duplication. The left-hand bar represents the size of chromosomes in megabases (Mb)

The amino acid (a.a) content of the *PvDDP* proteins varied between 585 a.a and 857 a.a, and their molecular weights (MWs) were found to be ranged from 66.5 kDa (*PvDDP7*) to 98.4 kDa (*PvDDP4*). Similar differences were observed in the pI values varying between 7.81 (*PvDDP10*) and 9.42 (*PvDDP9*). The subcellular localization of *PvDDPs* indicated that all the proteins were highly active on the plasma membrane of the common bean cells (Fig. 2). In silico tests on the stability of the proteins revealed that three out of 12 *PvDDPs* could be stable in the test tube, while all of them were estimated to be insoluble in *Escherichia coli* (Table 1). Amino acid compositions of the *PvDDPs* in common bean indicated that the most abundant amino acid found within these proteins is Leucine (Table S2).

Phylogenetic analysis of the *PvDDP* family members

In this study, to represent the evolutionary history of the *PvDDPs* and their relationships with each other was characterized by the phylogenetic analysis made

on the genes and the proteins structures. Amino acid sequence alignment carried out with Clustal W2 revealed that *PvDDPs* were characterized by fully conserved APEP amino acid motifs located on N-terminals of the domain (Fig. S1). *PvDDPs* in the common bean genome were divided into four categories according to the protein alignment. According to this classification, it was determined that *PvDDP1*, *PvDDP10*, *PvDDP11*, and *PvDDP12* were in the same group (Fig. 3). *PvDDP2*, *PvDDP7*, *PvDDP5*, and *PvDDP6* were observed to form another group. The third group includes *PvDDP4* and *PvDDP8*. The last group consists of two genes as *PvDDP3* and *PvDDP9*.

Conserved motif analysis revealed a total of 15 amino acid motifs were fully conserved in all *PvDDPs* except *PvDDP7*. Likewise, all the proteins, except *PvDDP7*, were predicted to have two transmembrane domains (RSN1_TM and RSN1_7TM) and a cytosolic domain of integral membrane protein (PHM7_cyt). Intron–exon structures of *PvDDP* genes are also very similar. Except for *PvDDP9* and

Table 1 Molecular and physicochemical features of *PvDDP* genes

Gene Name	Phytozome Identifier	Chromosome	Start and End positions (bp)	Length (bp)	CDS (pb)	Protein Length (A.A)	NCBI Accession number	Strand
<i>PvDDP01</i>	Phvul.001G031900.1.v2.1	chr01	2,905,945..2915984	2884	2256	751	XP_007160962.1	Forward
<i>PvDDP02</i>	Phvul.001G148800.1.v2.1	chr01	39,567,136..39576956	2718	2262	753	XP_007162399.1	Reverse
<i>PvDDP03</i>	Phvul.002G092500.1.v2.1	chr02	16,675,864..16684879	2869	2145	714	XP_007157718.1	Reverse
<i>PvDDP04</i>	Phvul.002G133000.1.v2.1	chr02	27,678,019..27684786	2980	2574	857	XP_007158205.1	Reverse
<i>PvDDP05</i>	Phvul.003G081500.1.v2.1	chr03	13,454,022..13460787	3138	2325	774	XP_007153987.1	Reverse
<i>PvDDP06</i>	Phvul.004G031700.1.v2.1	chr04	3,802,084..3814602	3100	2136	711	XP_007151261.1	Forward
<i>PvDDP07</i>	Phvul.005G030600.1.v2.1	chr05	2,831,287..2842993	2307	1758	585	XP_007148980.1	Reverse
<i>PvDDP08</i>	Phvul.006G000300.1.v2.1	chr06	160,081..166635	2915	2322	773	XP_007145950.1	Reverse
<i>PvDDP09</i>	Phvul.006G167700.1.v2.1	chr06	27,041,058..27045715	2783	2187	728	XP_007147944.1	Reverse
<i>PvDDP10</i>	Phvul.008G039200.1.v2.1	chr08	3,248,877..3257042	2446	2145	714	XP_007139547.1	Reverse
<i>PvDDP11</i>	Phvul.008G210700.1.v2.1	chr08	55,900,521..55907044	2792	2268	755	XP_007141612.1	Reverse
<i>PvDDP12</i>	Phvul.009G074900.1.v2.1	chr09	12,754,561..12766396	2476	2037	678	XP_007136794.1	Forward
Gene Name	pI	Protein Molecular Weight (Kda)	Instability index (II)	stable or unstable	GRAVY	Solubility/Score		
<i>PvDDP01</i>	8,8	85,11,155	44,84	Unstable	0,264	Insoluble/0.157		
<i>PvDDP02</i>	8,51	85,78,485	41,31	Unstable	0,128	Insoluble/0.153		
<i>PvDDP03</i>	9,23	81,8487	47,57	Unstable	0,287	Insoluble/0.127		
<i>PvDDP04</i>	9,16	98,04,348	58,81	Unstable	0,095	Insoluble/0.163		
<i>PvDDP05</i>	8,83	88,17,168	39,33	Stable	0,142	Insoluble/0.127		
<i>PvDDP06</i>	8,83	80,79,211	47,05	Unstable	0,199	Insoluble/0.142		
<i>PvDDP07</i>	8,66	66,50,528	41	Unstable	0,289	Insoluble/0.191		
<i>PvDDP08</i>	9,01	88,79,064	39,76	stable	0,16	Insoluble/0.152		
<i>PvDDP09</i>	9,42	81,96,123	32,77	stable	0,268	Insoluble/0.168		
<i>PvDDP10</i>	7,81	81,5393	48,61	Unstable	0,309	Insoluble/0.106		
<i>PvDDP11</i>	8,86	86,98,568	42,88	Unstable	0,152	Insoluble/0.211		
<i>PvDDP12</i>	8,54	77,09,412	46,5	Unstable	0,311	Insoluble/0.169		

PvDDP7, which have only five and six exons, all other *PvDDP* genes consist of 9–10 exons (Fig. 3). *PvDDPs*' evolution and their sequence similarities in other plant species were also investigated in the current study. For this purpose, a phylogenetic tree was constructed with the *DDPs* in common bean (*Pv*-12 genes), *Arabidopsis thaliana* (*At*-15 genes), *Glycine max* (*Gm*-22 genes), *Medicago truncatula* (*Mt*-13 genes), *Oryza sativa* (*Os*-11 genes), *Solanum lycopersicum* (*Sl*-12 genes), and *Zea mays* (*Zm*-14 genes) following the same workflow used in the identification of *PvDDP* genes (Fig. 4). The result of the phylogenetic analysis revealed nine groups, two of which

belong to only *SIDDP05* and *ZmDDP13*, representing their unique features compared to other *DDPs* in other plant species. *PvDDPs* were grouped into five different categories in the dendrogram with other plant species. Interestingly, four categories did not include any *PvDDPs* indicating the high similarities of *PvDDPs* between each other (Fig. 4).

In order to further study the potential evolutionary mechanisms of the *PvDDP* gene family, we identified two comparative syntenic maps of *P. vulgaris* with *Oryza sativa* and *Arabidopsis thaliana*, which were belong to monocotyledon and dicotyledon, respectively (Fig. 5). As a result, 14 collinear *PvDDP* gene

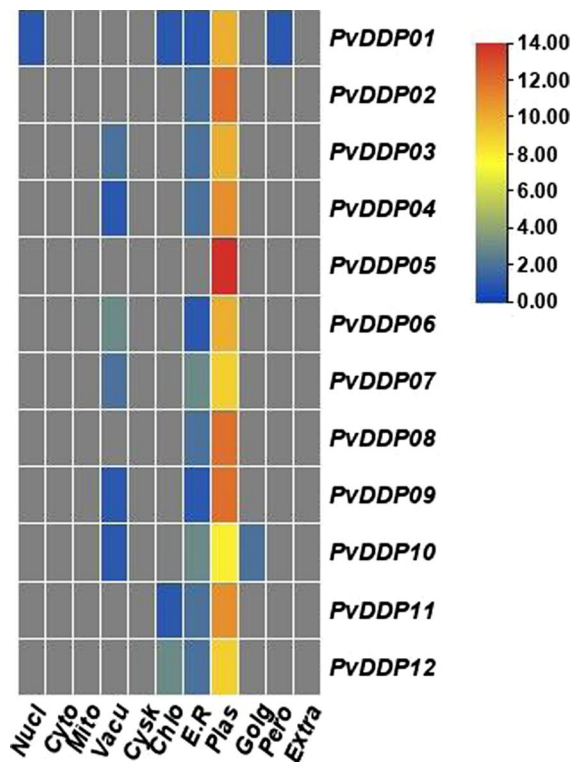


Fig. 2 Expression profile of *PvDDPs* in different tissues. WoLF PSORT web browser was used to generate the expression profiles of *PvDDPs* based on gene ontology and the UniProt database. The color scale at the right of the dendrogram indicates log2 expression values. Nucl, nucleus; Cyto, cytoplasm; Mito, mitochondria; Vacu, vacuole; Cysk, cytoskeleton; Chlo, chloroplast; E.R, endoplasmic reticulum; Plas, plasma membrane; Golg, Golgi apparatus; Pero, peroxisome; Extra, extracellular

pairs between *P. vulgaris* and *Arabidopsis thaliana* and five orthologs between *P. vulgaris* and *O. sativa* were introduced. The details of the collinear DUF gene pairs can be found in Table S3. The number of the orthologous events of *AtDDPs*-*PvDDPs* was more than that of *OsDDPs*-*PvDDPs*, indicating that of closer evolutionary distance between *P. vulgaris* and *A. thaliana*.

To clarify the evolutionary process of *PvDDP* genes, gene duplication events were also determined. Although some *PvDDPs* are found together on the same chromosomes (*PvDDP1*-*PvDDP2*; *PvDDP3*-*PvDDP4*; *PvDDP7*-*PvDDP8*; *PvDDP10*-*PvDDP11*), we could not identify any tandem duplication events. However, a total of four other types of duplication events were found. We identified

whole-genome duplication (WGD) events in pairs of *PvDDP1*-*PvDDP12* and *PvDDP6*-*PvDDP10*. In addition, it was determined that *PvDDP5*-*PvDDP8* and *PvDDP4*-*PvDDP11* pairs developed with DNA-transposed (DD) and dispersed duplication (DSD) events, respectively (Fig. 1).

Large-scale duplication may be dated using the consistent rate of per-site substitutions (Ks) that occur across time (Maher et al. 2006). The K_a , K_s , and K_a/K_s for each gene pair were determined to investigate the likely destiny of the *PvDDP* gene duplications. K_a/K_s was used as an indicator to decide selective pressure at sequence level among *PvDDP* genes; if this ratio equals zero, it indicates neutral selection. A ratio higher than one shows accelerated evolution with positive selection. Likewise, if this ratio is less than one, it is associated with negative selection (Juretic et al. 2005). Our in silico analysis revealed that K_a/K_s from four *PvDDP* duplicated gene pairs were less than 0.28 (Table 2). This result shows that the *PvDDP* gene family was primarily developed under high purifying selection pressure. About 3.9 to 18.4 million years ago (Mya), a duplication event is thought to have taken place, resulting in the creation of the two segmentally duplicated, one DNA-transposed, and one dispersed duplicated gene pair (Table 2).

Prediction of cis-elements, 3-D structures, and miRNAs targeting *PvDDP* genes

Light, hormone, and stress response-related cis-elements in the upstream region of 2000 nucleotides of *PvDDPs* from start codon were investigated to further clarify the functional role of *PvDDPs* in response to various stresses. This analysis revealed 45 different cis-regulatory elements in the putative promoter regions of the *PvDDPs* (Fig. 6). Among them, 12 cis-regulatory elements were predicted to be related to hormonal regulation, and the most abundant regulatory elements were ethylene-responsive element (ERE), ABA-responsive elements (ABRE), Jasmonate-responsiveness element (CGTA-motif), and SA-responsive elements (TCA-element). ERE and ABRE were found to exist in all *PvDDPs* indicating their potential regulation with ethylene and ABA hormones, both of which are strongly associated with biotic stress response in plant species. Jasmonate and SA-related cis-acting elements also represent

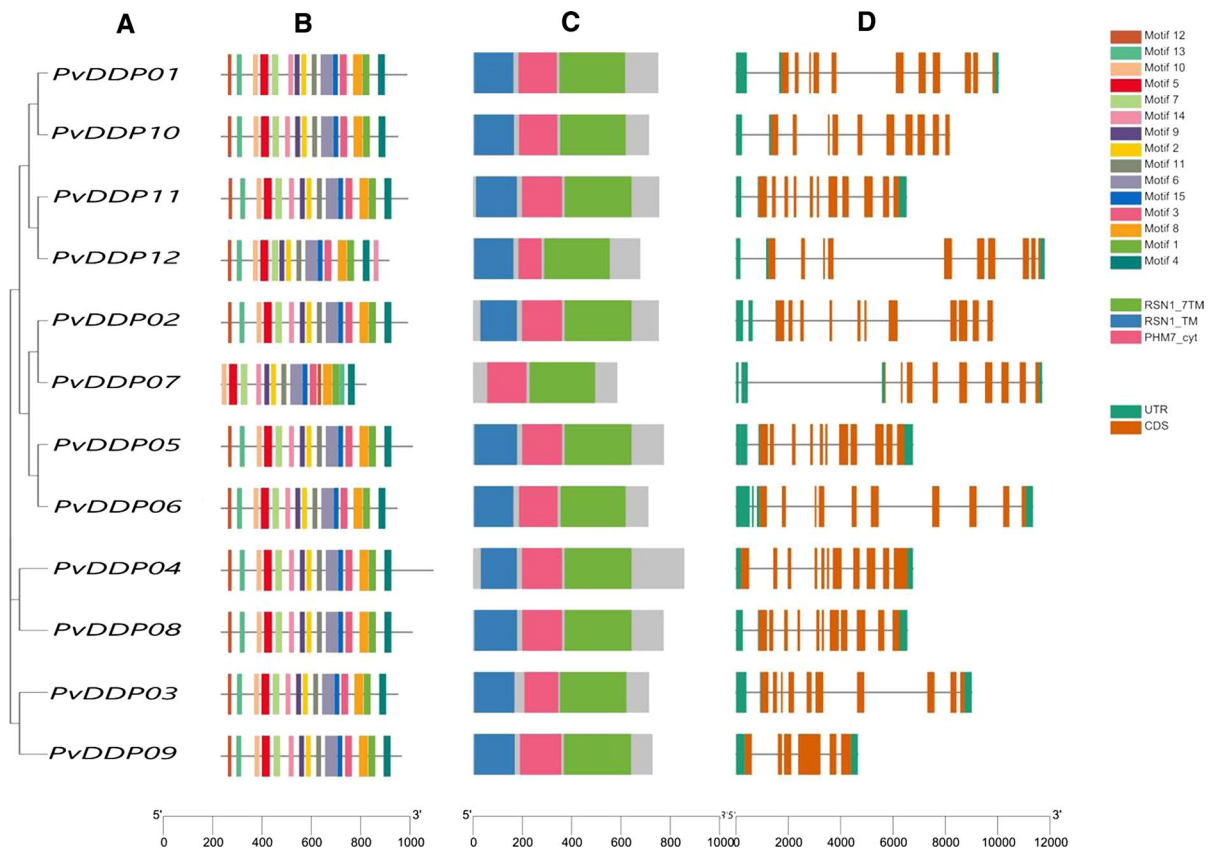


Fig. 3 Phylogenetic tree, gene structure conserved domains, and conserved motifs of 12 *PvDDPs*. **A** Phylogenetic tree was constructed through the maximum likelihood method based on the full-length protein sequences aligned with Clustal X 2.0. **B** Conserved motifs were provided through MEME Suite and illustrated in different boxes. **C** Distributed conserved domains

in *PvDDPs* with different colors. The green box indicates the RSN1_7TM domain. **D** Exon–intron structure of *P. vulgaris DDP* genes. Green boxes show untranslated 5' and 3' regions; brown boxes correspond to the exons. The sizes of the exons, introns, domains, and motifs can be estimated using the scale at the bottom

their potential contribution to abiotic stress response in the common bean. In addition to hormonal regulation, *PvDDPs* were also associated with various stress responses by the existence of 14 cis-regulatory regions in their promoter regions. As can be seen in Fig. 6, all the *PvDDPs* had abiotic stress-responsive cis-acting regulatory elements such as MYB and MYC in their promoter regions. Other abundant and common regulatory elements in the *PvDDPs* were antioxidant-responsive element (ARE), stress-response element (STRE) and pathogen-inducible cis-regulatory elements (AS-1). In the current study, the interaction between light and *PvDDPs* regulation in common bean was also investigated with in-silico estimation of the cis-acting elements in their promoter regions. This analysis also identified 14

different elements for the light response of *PvDDPs*. Among these cis-acting regulatory elements, BOX-4 and G-box were found to be common and abundant for all *PvDDPs*.

In silico modeling carried out by submitting protein sequences to Phyre2 server revealed that 12 *PvDDP* proteins had α -helix and β -sheet as their main structures (Fig. S2). However, it was observed that helix structures in all *PvDDP* proteins are quite numerous than beta-sheets. The tertiary structure of these proteins was clearly found to be quite similar, supporting the annotation in PDB that they were transport proteins. This result was supported by analysis with TMHMM—2.0, which showed that all *PvDDP* proteins have transmembrane helices (Fig. S3).

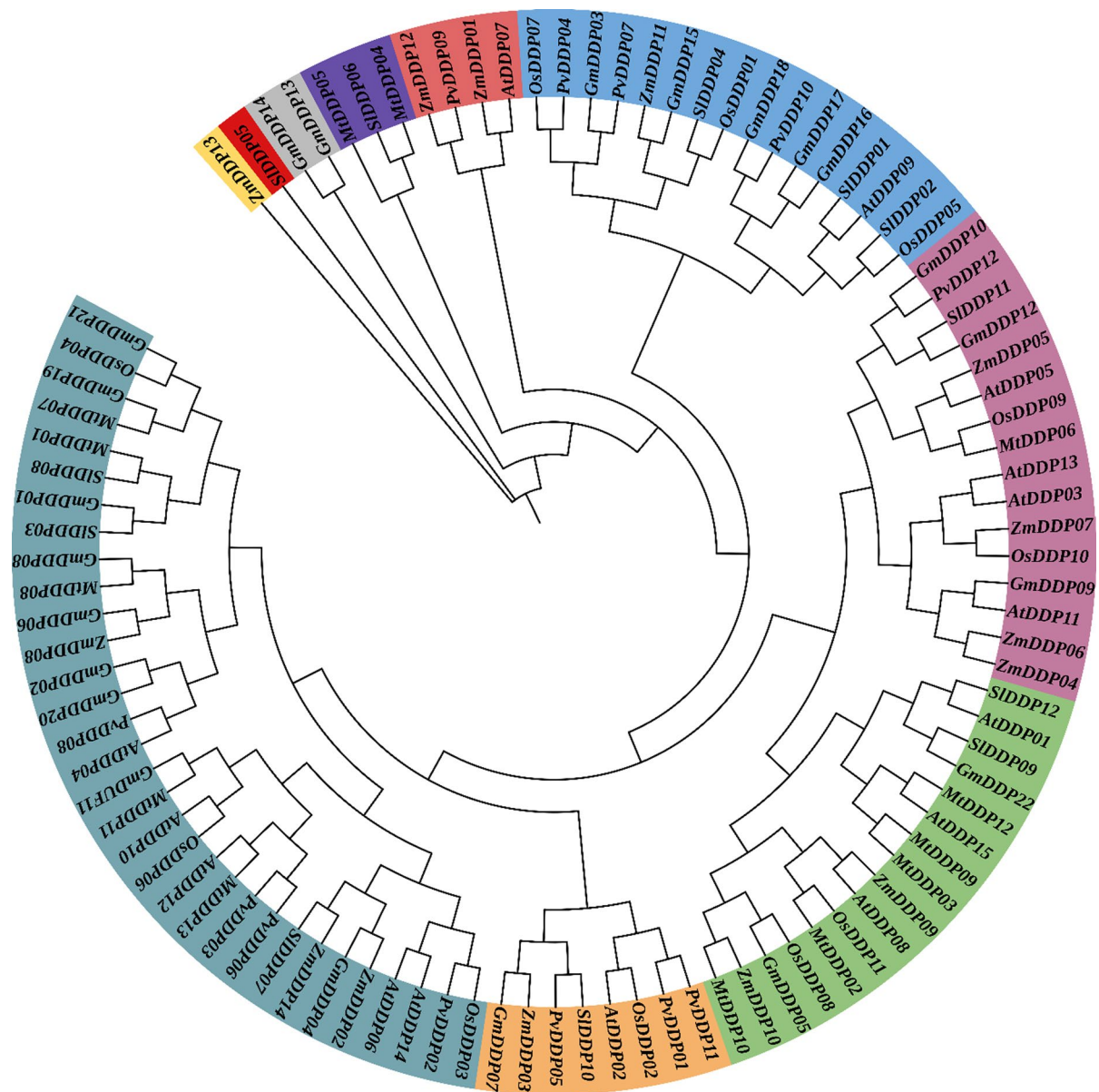


Fig. 4 Phylogenetic relationship of PDD genes from common bean, *Arabidopsis thaliana*, *Glycine max*, *Cucumis sativus*, *Medicago truncatula*, *Solanum lycopersicum* L. and *Oryza sativa*. PDD genes are divided into ten subfamilies. The tree

was created through the MEGA X software by the maximum likelihood method with a bootstrap of 1000 replicates and visualized with ITOL v3

Regulation of the genes or transcription factors depends not only on the regulatory elements of the gene' promoters but also on miRNA interference in plant cells. Therefore, in the current study, 16 miRNAs potentially targeting the 9 *PvDDPs* were identified according to their complementarity

to the transcript sequences (Table S4). *PvDDP3*, *PvDDP6*, and *PvDDP9* had no miRNA complementarity required for targeting their transcripts. On the other hand, the *PvDDP4*, *PvDDP5*, and *PvDDP8* had one miRNA, while the others had two miRNAs. The highest number of miRNAs were identified for *PvDDP10* (4 miRNAs).

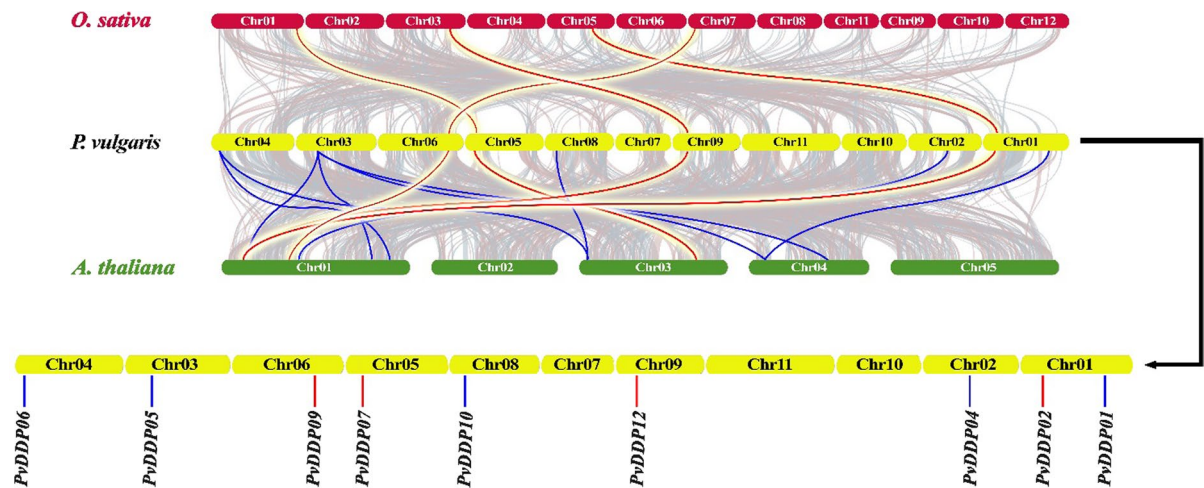


Fig. 5 Synteny analysis and gene duplication of *DDP* genes between common bean and both plants. Each species is given in different colors. Red lines in the background indicated the syntenic *DDP* gene pairs between *A. thaliana*, *O. sativa*, and *P. vulgaris*, but blue lines in the background indicated the syn-

tenic *DDP* gene pairs between *A. thaliana* and *P. vulgaris*. Gray lines in the background showed the collinear blocks between species. At the bottom, locations of *DDP* genes were illustrated in the *P. vulgaris* chromosomes

Table 2 Ka/Ks of *PvDDP* genes

Gene 1	Gene 2	Identity (%)	Ka	Ks	Ka/Ks	Duplicated Type	Purifying selection	Effective Length	Mya
<i>PvDDP01</i>	<i>PvDDP12</i>	65	0.16758367	0.634325838	0.264191777	WGD	Yes	2028	4.879429526
<i>PvDDP5</i>	<i>PvDDP8</i>	84	0.090043762	0.579639933	0.155344304	DD	Yes	2310	4.458768719
<i>PvDDP6</i>	<i>PvDDP010</i>	74	0.143926397	0.507178674	0.283778488	WGD	Yes	2124	3.901374419
<i>PvDDP04</i>	<i>PvDDP11</i>	64	0.317619924	2.402396284	0.13220963	DSD	Yes	2253	18.47997142

WGD: whole-genome duplication

DD: DNA-transposed

DSD:dispersed duplication

Pvu-miRN2594, Pvu-miR169, Pvu-miRN2584, and Pvu-miR530 are predicted to target *PvDDP10*, while Pvu-miR156 and Pvu-miRN2592 are predicted to target the *PvDDP11* gene.

Protein–protein interaction analysis

Selecting those *PvDDPs* with a confidence score higher than 0.7, seven functional-enriched networks (with different labels) and ten functional partners (XP_007137711.1, XP_007141321.1, XP_007135877.1, XP_007155772.1, XP_007161051.1, XP_007144372.1, XP_007145923.1, XP_007157224.1,

XP_007160071.1, XP_007141722.1) were introduced by means of the following databases in Fig. 7: Through STRING database, two groups were identified; one with turkey blue label associated with late exocytosis and golgi transport (DDP7, XP_007145923.1, and XP_007141321.1), and the other labeled with purple related to tRNA uridine (34) hydroxylase, and late exocytosis (XP_007137711.1, DDP4, DDP5, DDP10, and DDP11). Utilizing the KEGG pathway, other two groups were introduced; one is labeled as green associated with metabolic pathways (XP_007149185.1, XP_007137006.1, XP_007160071.1, XP_007143591.1, XP_007145616.1, XP_007136748.1,

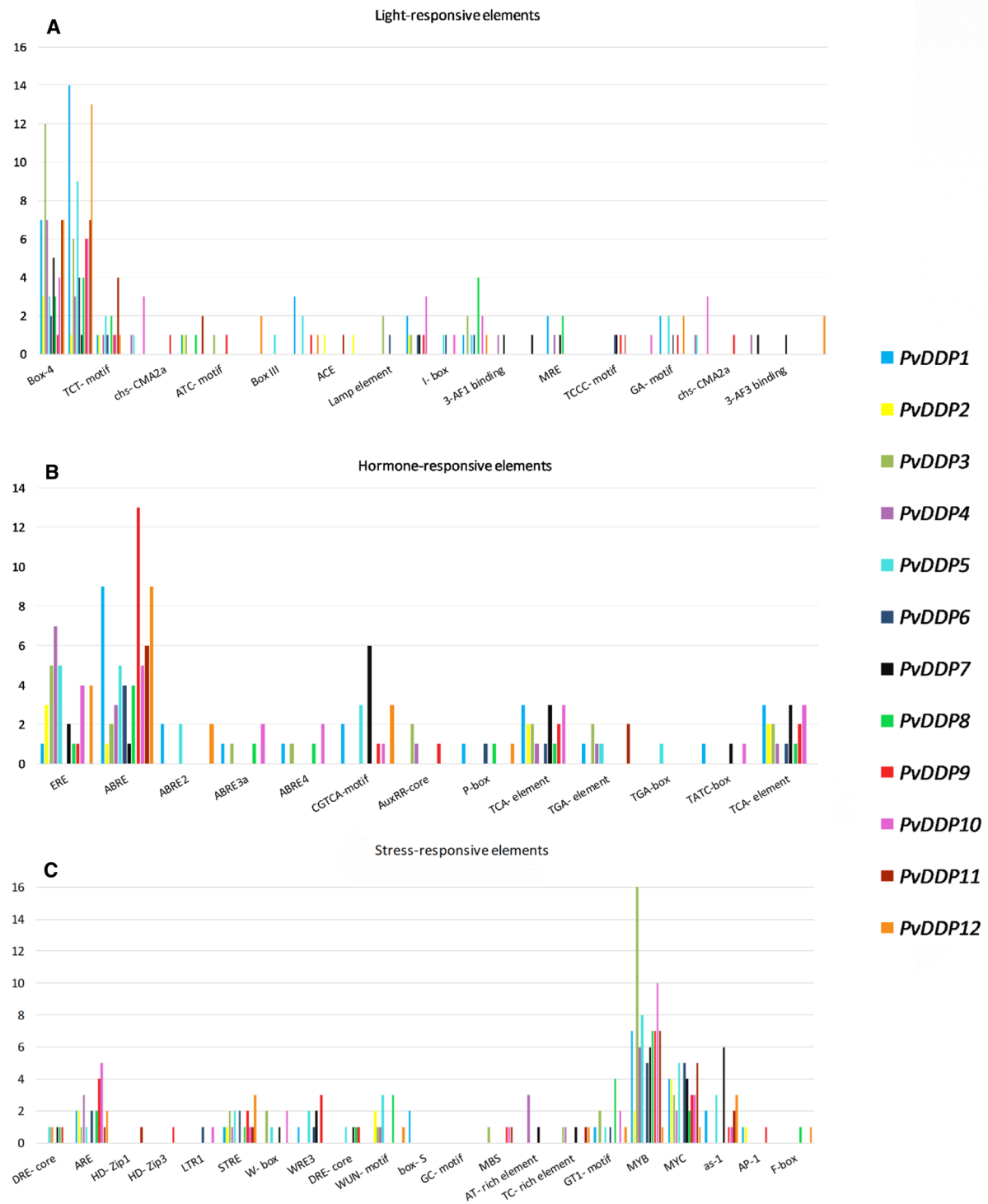


Fig. 6 Stress (a), hormone (b), and light (c) response related cis-elements in the promoter sequences of *PvDDPs* covering the upstream region of 2000 nucleotides from the start codon

of gene. Different colors are representing different *PvDDPs*. The families of cis-elements were described through the PlantCARE database

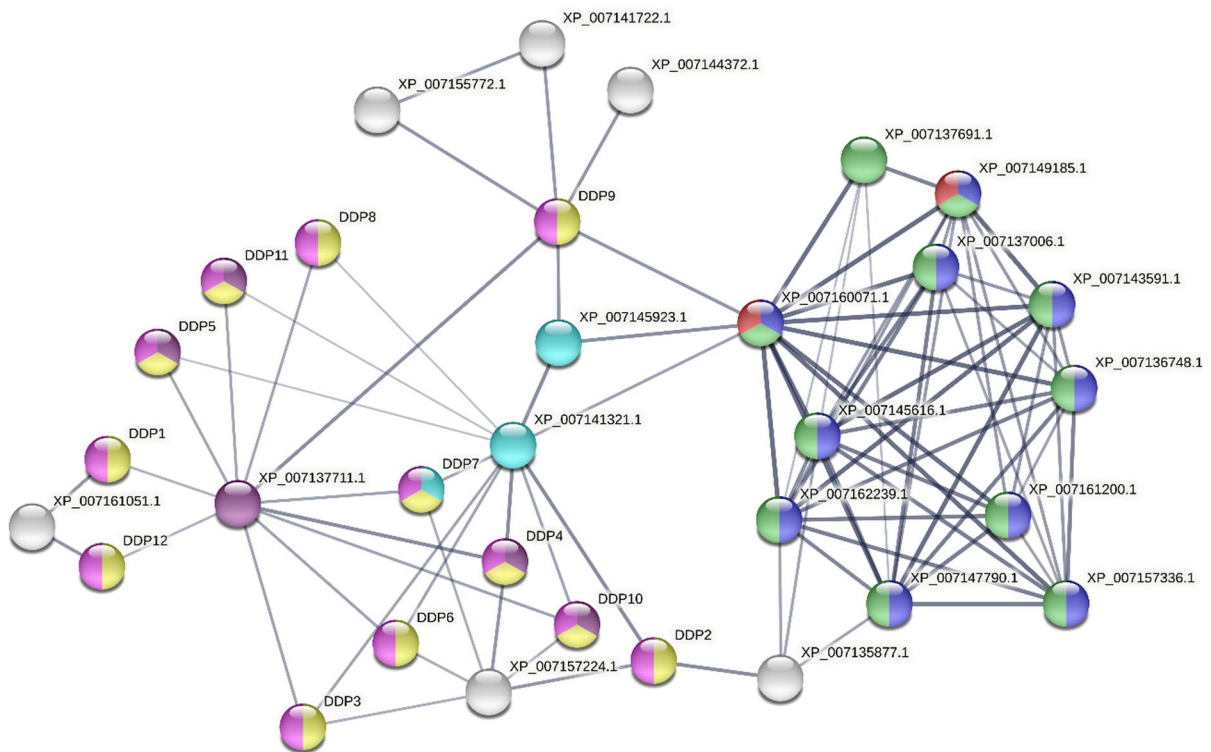


Fig. 7 The protein–protein interaction (PPI) network of DDPs was built by the STRING database in *Phaseolus vulgaris*. The thickness of the line indicates the strength of interaction. Based on the enriched network cluster, seven enriched networks were generated in different colors: turkey blue and purple color enriched network originate from STRING database; dark blue

and green color enriched network root in KEGG pathway database; red color enriched network is derived from UniProt database; yellow color enriched network arise from Pfam database; and pink color enriched network originate from InterPro database

XP_007162239.1, XP_007161200.1, XP_007147790.1, XP_007157336.1, and XP_007137691.1), and the other labeled as a dark blue link to purine metabolism (same protein within metabolic pathways except for XP_007137691.1). The UniProt database identified one group labeled as red associated with GMP biosynthesis (XP_007160071.1 and XP_007149185.1). Based on Pfam and InterPro database, all *PvDDPs* were found to be associated with calcium-permeable stress-gated cation channel 1, N-terminal transmembrane domain as marked yellow and associated with Calcium-dependent channel, 7TM region, putative phosphate as marked pink, respectively. All details about matching protein IDs and labels in the network and strength, false discovery rate, and scores can be found in Table S5.

Stress-responsive expression profiles of DUFs

Expression patterns of *PvDDPs* in different stress conditions were also evaluated with available RNA-seq data sets retrieved from the NCBI SRA database. Six transcriptome data were analyzed to explore the regulation of the *PvDDPs* in common bean under drought, salt, and pathogen stress. Their expression ($\log_2$ fold change) was quantified in a heat map to represent their regulation under these stressors. As shown in Fig. 8, $\log_2$ FC values of *PvDDPs*.

were determined to be between -2 and $+2.5$ under different abiotic and biotic stress conditions. In the first transcriptome comparison, where leaves of salt-tolerant Ispir genotype and sensitive T43 genotype were compared under salt-stress conditions, it was determined that the expression of *PvDDP* 1–3–4–5–7–9–13 genes was decreased. Other *PvDDPs* were not differentially expressed in the same transcriptome

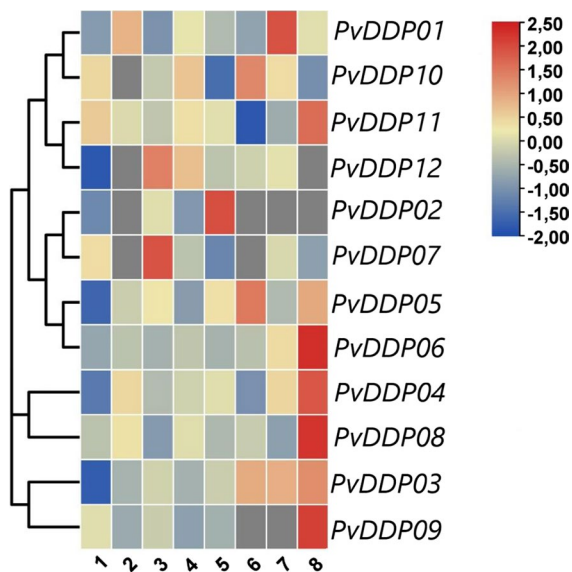


Fig. 8 Heat map showing expression of *PvDDPs* in various RNA-seq datasets. Blue, negative log fold-change (log FC) refers lower expression in the first genotypes compared with the second; red, positive log FC. The raw RNA-seq datasets were acquired from NCBI Sequence Read Archive (SRA) under the accession number; 1 and 2 indicate salt-treated leaf explants of T43 (sensitive) and Ispir (resistant) common bean genotypes (PRJNA656794), 3 indicates root tissues of salt-tolerant Ispir grown under salt stress and control conditions (PRJNA656794), 4 represents salt-treated tolerant and sensitive genotypes at the bud stage (PRJNA558376), 5 indicates salt-treated tolerant bean genotype at the sprout stage (PRJNA691982), 6 and 7 indicate RNA sequencing of drought-tolerant *Perola* (PRJNA327176) and *Pinto saltillo* (PRJNA508605) genotypes grown under drought stress, and control conditions and 8 represents a fungal pathogen *Sclerotinia sclerotiorum* (strain 1980) infected common bean (PRJNA574280). CyVerse and Galaxy (usegalaxy.eu) tools were used to analyze the transcription of downloaded data. HISAT2 tool was conducted to map of reads. Stringtie 1.3.3 and Ballgown were performed for Transcript assembly and differential expression analyses, respectively. Differentially expressed genes (DEGs) represent the genes having fold change value $\log_2 > 1$ and a p -value < 0.05 . Heatmap based on the $\log_2FC$ was illustrated via TBTools

comparison. The same transcriptome comparison in the salt-treated roots indicated non-significant differential expression for all *PvDDPs* (2nd transcriptome data- Fig. 8). In the third transcriptome comparison, the salt-treated leaves of the salt-tolerant Ispir genotype were compared to its control counterpart. The results indicated that only two members, *PvDDP1* and *PvDDP13*, were differentially upregulated and repressed respectively in salt-treated leaves of Ispir

(3rd transcriptome data- 8). Like in roots, *PvDDPs* were not differentially expressed in salt-treated buds in tolerant Ispir compared to sensitive T43 (4th transcriptome data- Fig. 8). The results of transcriptome comparison between salt-treated sprout of T43 and Ispir indicated differential expression only for two.

PvDDPs in tolerant genotype compared to sensitive one; repression of *PvDDP2* and induction of *PvDDP1* (5th transcriptome data- Fig. 8). Drought treatment upregulated three *PvDDPs* (6–8–10) in tolerant common bean genotype (Pinto Saltillo) in the 6th transcriptome data (Fig. 6), while the same stress did not affect the expression of any *PvDDPs* in the other two genotypes (Perola and BAT477) (7th transcriptome data- Fig. 8). The most significant differential regulation was recorded for the fungal pathogen-treated leaves of the common bean genotype. *PvDDP3*–4–5–7–0 and 12 significantly upregulated under biotic stress, whereas *PvDDP11* and *PvDDP6* were downregulated with pathogen infection.

Determination of expression pattern of DUFs under various stress conditions

The expression levels of *PvDDP* genes were measured on the salt, drought, IAA, and ABA treated roots and leaves of common bean plants by quantitative real-time PCR. Under salt stress, all the *PvDDPs* were significantly up-regulated, especially under prolonged stress conditions. *PvDDP5*–9 and *PvDDP10* were revealed similar regulation patterns in response to salt stress in the leaves and roots. Their expression level increased at the first 6 h more in salt-treated roots and then reduced with the acclimation of the plant to the stress condition. However, their expression levels were again upregulated in leaves and roots with prolonged salt stress in the common bean. The *PvDDP1* was the fastest responsive protein due to its up-regulation at the first stage, but its expression was reduced with an increase in salt stress. *PvDDP2*–3–4–6–8–11 and 12 indicated the same expression pattern with salt stress. Their expression levels were increased with prolonged salt stress (7 days), mostly in the leaf samples (Fig. 9). The up-regulation of the *PvDDPs* were ranged from >2 and 470 folds in prolonged salt-treated leaves, and their expression level were ordered as *PvDDP11* $>$ 2 $>$ 8 $>$ 12 $>$ 3, respectively. Contrary to higher expression of the *PvDDPs* in the salt-treated common bean leaves, expression

of all the *PvDDPs* upregulated on drought-treated root tissue. Similar to salt treatment, the expression level of *PvDDP1* was increased at the 6th hour of PEG treatment and reduced with an increase in drought treatment. Interestingly, all other *PvDDPs* were strongly upregulated only on the 4th day of the PEG treatment in common bean roots. The expression increase of *PvDDPs* was suddenly increased on the 4th day, and upregulation of the genes was varied from 3.2 folds to 1500 folds (*PvDDP11* > 2 > 8 > 6 > 9 > 10 > 12) in PEG treated common bean roots. Despite strong upregulation of the genes on the 4th day of PEG treatment, the expression level was not continued through the end of stress and reduced on the 7th day of the treatment.

Abscisic acid (ABA) is a plant hormone that regulates numerous aspects of plant growth, development, and stress responses. Therefore, the interaction and correlation of the *PvDDPs* with ABA was investigated in the current study to understand their potential roles in common beans. Compared to the salt and PEG treatment, *PvDDPs* have indicated much smaller upregulation in response to ABA treatment in the leaves and roots of common bean. *PvDDP 3–4–5–6–7–10* and *12* indicated less than three-fold expression increase in common bean tissues. Like in salt treatment, the highest expression was recorded in *PvDDP11* after 60 h in the.

leaves subjected to ABA. Expression of *PvDDP2* and *PvDDP8* was also increased at the same treatment hour in the same tissue. Due to its vital role in plant growth regulation and developmental processes, IAA (auxin) was also applied to the common bean plant to evaluate its interaction with *PvDDPs* expression. Like ABA, the expression level of some *PvDDPs* (3–4–5–6–7–9–10) was induced less than three folds in IAA-treated leaves and roots. Like all other treatments, *PvDDP11* was the highest upregulated (> 50 folds) gene on the 60th hour for only IAA-treated leaves. Although expression levels of the *PvDDP1–2–4–5–6–7–10–12* indicated a slight positive induction at the beginning of IAA treatment, especially in the roots, this weak upregulation was reduced with an increase in hormone treatment.

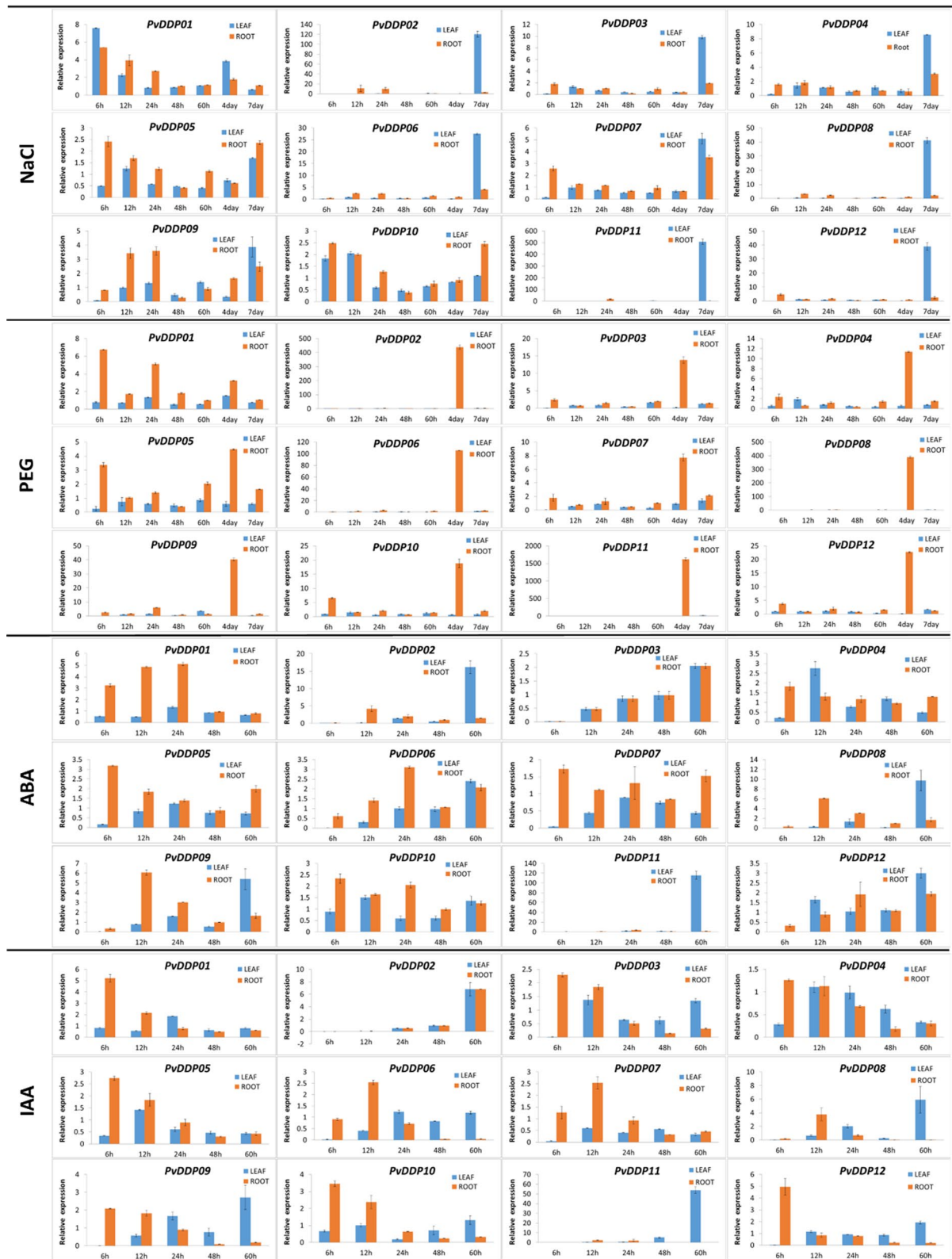
Overall, as evident from these results, *PvDDPs* are not much regulated at the onset of the hormone treatment compared to drought and salt stress. It is clearly seen that there is a strong relationship between abiotic stress and *PvDDPs* expression in both leaf and

root tissues. *PvDDP11* is overtly the highest upregulated gene with hormone treatment and abiotic stress among all the members. Expression of the *PvDDP11* gene indicated a strong correlation with PEG and NaCl treatment in the common bean roots and leaves, respectively.

Discussion

The largest family set in protein databases (Pfam) is domains of unknown function (DUFs), which are frequently used to identify any protein with an uncertain function. The number of DUF families has increased in recent years due to the development of next-generation sequencing systems and the increase in the number of genomes sequenced in parallel. As a result of these developments, the number of DUF gene families in the PFAM database has increased rapidly in the last ten years (Yang et al. 2020). Their large number in databases can complicate the determination of the functions of these proteins. However, genome-wide analyses of these gene families in different plants and expression analyses performed under different conditions will facilitate understanding the functions of these gene families. In order to contribute to the understanding of this obscurity, we tried to identify the members of the DUF gene family in the common bean plant and analyze their expression after ABA, IAA, salt, and drought stress applications.

In this study, 12 members of the DUF221 family, characterized by the RSN1_7TM domain, were identified using in silico tools. In fact, although we found a total of 13 genes as a result of Phytozome analyzes, as a result of the extensive domain analyzes we performed later, we found that one gene did not meet all the criteria. Since it is the first study in the literature analyzing the DUF221 domain in plant genomes, it cannot be interpreted with the numbers in other plants, but other DUF families are found in similar numbers in different plants. For example, as a result of the study carried out to determine the members of the gene family containing the DUF966 domain in the wheat plant, a total of 28 genes were identified (Zhou et al. 2020). In the analysis performed by Zaho et al. in the cotton plant in 2021, they determined the number of genes containing the DUF668 domain as 32. Although there are different members of the DUF family analyzed in these studies, it can be said that



◀Fig. 9 qRT-PCR analysis of *PvDDPs* in leaves and roots of common bean plants under drought, salt, ABA, and IAA treatments. Expression levels were conducted at different time points. Based on the $2^{-\Delta\Delta C_t}$ method, relative expression was measured

the main difference in gene numbers is related to the genome size of the analyzed plant. We can also say such differences between DUF domain families in terms of intron–exon numbers. While the number of exons of the genes containing the DUF221 domain in the common bean plant varies between 5 and 10, in another study conducted in *Arabidopsis thaliana*, it was reported that the majority of the genes belonging to the DUF4228 family were without introns, while the other genes had 2–3 introns (Yang et al. 2020). We performed phylogenetic analyzes using *DDP* genes from common beans and six other organisms. As a result of this neighbor-joining-based analysis, we determined that *DDP* genes formed nine different groups. These results we obtained are compatible with the study in which genome-wide analysis of DUF4228 genes in soybean was performed. Because in that study, the members of the DUF4228 family, which are found in *Arabidopsis*, soybean, and *Medicago truncatula* plants, were subjected to phylogenetic analysis, and it was reported that eight different subgroups were formed (Leng et al. 2021).

We used the PPI network to understand the function of *PvDDPs* further. In the PPI network, *PvDDPs* do not have direct interactions among themselves. However, all *PvDDPs*, except for *PvDDP2*, are indirectly connected through XP_007137711.1, which has PME1 domain invertases, forming a non-covalent 1:1 complex (PubMed:8,521,860) and has been involved in the regulation of carbohydrate metabolism, cell wall extension, and fruit development. The other important functional enriched prediction in the network is the connection between *PvDDPs* and purine metabolism (blue label), metabolic pathways (green label), and GMP biosynthesis (red label) through *PvDDP9*, XP_007145923.1, and XP_007141321.1 with a high confidence score. GMP biosynthesis is part of purine metabolism regarded as a housekeeping activity on nitrogen remobilization for plant growth and development. However, recent evidence indicates that certain purine metabolites attribute to plant stress responses and stress adaptations. For example, several plants respond to biotic

and abiotic stress activating enzymes in the purine degradation pathway (Alamillo et al. 2010; Hesberg et al. 2004; Montalbini 1995; Sagi et al. 1998; Watanabe et al. 2014). The high score of *PvDDPs* and their associated functional proteins involved in purine metabolism strengthens the hypothesis that *PvDDPs* can be associated with abiotic stress response and signaling. Although not provided by functional enrichment, the functional interaction between *PvDDP9*, XP_007155772.1 (Short name is LEA_1), and XP_007141722.1 (PAM2-containing protein CID1/CID2), whose high confidence score and direct connections, indicate that all *PvDDPs*, especially *PvDDP9*, can be involved in abiotic stress response and hormone signaling. Because Pfam LEA_1 entry is involved in LEA6, LEA18, and LEA46 from *A. thaliana* and those LEA proteins play essential roles in the adaptive process against water deficit (Olvera Carrillo et al. 2010). Moreover, PAM2-containing protein CID1/CID2, also known as polyadenylate-binding protein-interacting protein 1 or protein EARLY RESPONSIVE TO DEHYDRATION 15 (ERD15), is a negative regulator of ABA (abscisic acid) responses (Kariola et al. 2006). ERD15 is included in stress tolerance in plants, involving tolerance to drought (Aalto et al. 2012; Shao et al. 2014).

Expression profiles of *PvDDPs* were evaluated not only RNA-seq based data but also by using qRT-PCR analysis with tissues applied to different stress conditions. Six transcriptome data sets were analyzed to determine how the *PvDDPs* in the common bean are regulated in response to drought, salt, and pathogen stress. Although all *PvDDP* genes annotated as transport proteins, under salt stress, only *PvDDP1* and drought stress, three *PvDDPs* (6–8–10) increased differently, while the most significant increase was in fungal pathogen-treated leaves of common bean genotype a total of 6 *PvDDP* genes (3–4–5–7–10 and 12) increased differently. These results are consistent with previous studies in the literature. For example, in a study of genome-wide analysis of the OsDUF668 family in rice plants, they determined that OsDUF668-1, -3, -4, and -5 were significantly elevated after wound and rice blasting conditions. They reported that transcripts of these genes were particularly strongly induced by rice blast (Zhong et al. 2019). Similarly, our study showed less differentially expressed genes under salt stress, while more genes were active under pathogen stress.

Plant growth regulators in plants, including auxin, cytokinin, and ethylene, have a long-term effect on the structure and growth of plants (Marciniak Przedniczek 2019). We employed qRT-PCR to examine the relative expressions of *PvDDPs* under IAA and ABA treatments in order to investigate their potential activities in response to hormone stress. We observed that *PvDDPs* were not very effective at the start of both ABA and IAA hormone treatment. By qRT-PCR analysis, we detected a strong correlation between drought and salt stress and *PvDDP* expression in both leaf and root tissues. Because DUF genes act as cytosolic defense proteins and it has been suggested that they exert defense responses to the cell wall during stress conditions in plants (Gholizadeh 2020). We also found that the *PvDDP11* gene is highly expressed in salt and drought conditions. This situation shows great consistency with miRNA analysis because the *PvDDP11* gene is targeted by miRNA156 and Pvu-miR2592, which are associated with the abiotic stress response and growth regulation (Qiu et al. 2019; Wang et al. 2011).

Conclusion

This is the first time that the common bean DDP family (DUF221) has been studied in-depth, and the link between this family and drought, salt, IAA, and ABA stress response was assessed. This study served as a valuable resource for future research on the structure and function of DDP proteins in the regulation of drought, salt, IAA, and ABA stress responses in plants.

Author Contributions MK planned, designed, and performed the experiments, KA, ZS, BAY and GG analyzed the data, wrote the paper; KY wrote the paper. All authors have read and approved the final manuscript.

Funding This study was supported by the Research Fund of Ondokuz Mayıs University (PYO.ZRT.1901.17.010).

Data availability The datasets analysed during the current study are available in the NCBI/SRA. Database with following accession numbers; PRJNA508605 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA508605>), PRJNA327176 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA327176>), PRJNA574280 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA574280>), PRJNA691982 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA691982>), PRJNA656794 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA656794>), PRJNA558376 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA558376>), and PRJNA574280 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA574280>).

[ncbi.nlm.nih.gov/bioproject/PRJNA656794](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA656794)), PRJNA558376 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA558376>), and PRJNA574280 (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA574280>).

Declarations

Conflict of interest The authors declare that they have no conflict of interests. The funding sponsors had no role in the design of the study, collection, analyses, interpretation of the data, writing of the manuscript, and in the decision to publish the results.

References

- Aalto MK, Helenius E, Kariola T, Pennanen V, Heino P, Hörak H, Puzõrjova I, Kollist H, Palva ET (2012) ERD15—An attenuator of plant ABA responses and stomatal aperture. *Plant Sci* 182:19–28. <https://doi.org/10.1016/j.plantsci.2011.08.009>
- Alamillo JM, Díaz-leal JL, Sánchez-moran MV, Pineda M (2010) Molecular analysis of ureide accumulation under drought stress in *Phaseolus vulgaris* L. *Plant Cell Environ* 33:1828–1837. <https://doi.org/10.1111/j.1365-3040.2010.02187.x>
- Bateman A, Coghill P, Finn RD (2010) DUFs: families in search of function. *Acta Cryst* 66:1148–1152. <https://doi.org/10.1107/S1744309110001685>
- Bayuelo Jimenez J, Jasso Plata N, Ochoa I (2012) Growth and physiological responses of *Phaseolus* species to salinity stress. *Int J Agron*. <https://doi.org/10.1155/2012/5276731>
- Broughton WJ, Hernandez G, Blair M, Beebe S, Gepts P, Vanderleyden J (2003) Beans (*Phaseolus* spp.)—model food legumes. *Plant Soil* 252:55–128. <https://doi.org/10.1023/A:10241467106111>
- Cao X, Yang KZ, Xia C, Zhang XQ, Chen LQ, Ye D (2009) Characterization of DUF724 gene family in *Arabidopsis thaliana*. *Plant Mol Biol* 72:61. <https://doi.org/10.1007/s11103-009-9551-5>
- Chen C, Chen H, He Y, Xia R (2018) TBtools, a toolkit for biologists integrating various biological data handling tools with a user-friendly interface. *BioRxiv*:289660. Doi: <https://doi.org/10.1101/289660>
- Chou KC, Shen HB (2008) Cell-PLoc: a package of web servers for predicting subcellular localization of proteins in various organisms. *Nat Protoc* 3:153. <https://doi.org/10.1038/nprot.2007.494>
- Cortés AJ (2013) On the Origin of the Common Bean (*Phaseolus vulgaris* L.). *Am J Plant Sci* 4 (10):1998–2000. <https://doi.org/10.4236/ajps.2013.410248>
- Dai X, Zhuang Z, Zhao PX (2018) psRNATarget: a plant small RNA target analysis server (2017 release). *Nucleic Acids Res* 46:W49–W54. <https://doi.org/10.1093/nar/gky316>
- Enright A, John B, Gaul U, Tuschl T, Sander C, Marks D (2003) MicroRNA targets in *Drosophila*. *Genome Biol* 4:1–27. <https://doi.org/10.1186/gb-2003-4-11-p81>
- Finn RD, Clements J, Arndt W, Miller BL, Wheeler TJ, Schreiber F, Bateman A, Eddy SR (2015) HMMER web

- server: 2015 update. *Nucleic Acids Res* 43:W30–W38. <https://doi.org/10.1093/nar/gkv3971>
- Finn RD, Coghill P, Eberhardt RY, Eddy SR, Mistry J, Mitchell AL, Potter SC, Punta M, Qureshi M, Sangrador Vegas A (2016) The Pfam protein families database: towards a more sustainable future. *Nucleic Acids Res* 44:D279–D285. <https://doi.org/10.1093/nar/gkv13441>
- Ganie SA, Pani DR, Mondal TK (2017) Genome-wide analysis of DUF221 domain-containing gene family in *Oryza* species and identification of its salinity stress-responsive members in rice. *PLoS ONE* 12:e0182469. <https://doi.org/10.1371/journal.pone.01824691>
- Gholizadeh A (2020) Pectin methylesterase activity of plant DUF538 protein superfamily. *Physiol Mol Biol Plants* 26:829–839. <https://doi.org/10.1007/s12298-020-00763-91>
- Guo Z, Kuang Z, Wang Y, Zhao Y, Tao Y, Cheng C, Yang J, Lu X, Hao C, Wang T (2020) PmiREN: a comprehensive encyclopedia of plant miRNAs. *Nucleic Acids Res* 48:D1114–D1121. <https://doi.org/10.1093/nar/gkz8941>
- Hesberg C, Hänsch R, Mendel RR, Bittner F (2004) Tandem orientation of duplicated xanthine dehydrogenase genes from *Arabidopsis thaliana*: differential gene expression and enzyme activities. *J Biol Chem* 279:13547–13554. <https://doi.org/10.1074/jbc.M3129292001>
- Hiz MC, Canher B, Niron H, Turet M (2014) Transcriptome analysis of salt tolerant common bean (*Phaseolus vulgaris* L.) under saline conditions. *PLoS ONE* 9:e92598. <https://doi.org/10.1371/journal.pone.00925981>
- Hon J, Marusiak M, Martinek T, Kunka A, Zendulka J, Bednar D, Damborsky J (2021) SoluProt: prediction of soluble protein expression in *Escherichia coli*. *Bioinform* 37:23–28. <https://doi.org/10.1093/bioinformatics/btaa11021>
- Hou C, Tian W, Kleist T, He K, Garcia V, Bai F, Hao Y, Luan S, Li L (2014) DUF221 proteins are a family of osmosensitive calcium-permeable cation channels conserved across eukaryotes. *Cell Res* 24:632–635. <https://doi.org/10.1038/cr.2014.141>
- Jamil A, Riaz S, Ashraf M, Foolad MR (2011) Gene expression profiling of plants under salt stress. *Crit Rev Plant Sci* 30:435–458. <https://doi.org/10.1080/07352689.2011.6057391>
- Juretic N, Hoen DR, Huynh ML, Harrison PM, Bureau TE (2005) The evolutionary fate of MULE-mediated duplications of host gene fragments in rice. *Genome Res* 15:1292–1297. <https://doi.org/10.1101/gr.40642051>
- Kariola T, Brader G, Helenius E, Li J, Heino P, Palva ET (2006) EARLY RESPONSIVE TO DEHYDRATION 15, a negative regulator of abscisic acid responses in arabidopsis. *Plant Physiol* 142:1559–1573. <https://doi.org/10.1104/pp.106.0862231>
- Kavas M, Baloglu MC, Atabay ES, Ziplar UT, Daşgan HY, Ünver T (2016) Genome-wide characterization and expression analysis of common bean bHLH transcription factors in response to excess salt concentration. *Mol Genet Genom* 291:129–143. <https://doi.org/10.1007/s00438-015-1095-61>
- Kavas M, Kızıldoğan AK, Abanoz B (2017) Comparative genome-wide phylogenetic and expression analysis of SBP genes from potato (*Solanum tuberosum*). *Comput Biol Chem* 67:131–140. <https://doi.org/10.1016/j.compbiolchem.2017.01.0011>
- Kavas M, Yildirim K, Secgin Z, Abdulla MF, Gokdemir G (2021) Genome-wide identification of the BURP domain-containing genes in *Phaseolus vulgaris*. *Physiol Mol Biol Plants* 27:1885–1902. <https://doi.org/10.1007/s12298-021-01052-91>
- Kelley LA, Mezulis S, Yates CM, Wass MN, Sternberg MJ (2015) The Phyre2 web portal for protein modeling, prediction and analysis. *Nature Protoc* 10:845–858. <https://doi.org/10.1038/nprot.2015.0531>
- Krogh A, Larsson B, Von Heijne G, Sonnhammer EL (2001) Predicting transmembrane protein topology with a hidden markov model: application to complete genomes. *J Mol Biol* 305:567–580. <https://doi.org/10.1006/jmbi.2000.43151>
- Leng ZX, Liu Y, Chen ZY, Guo J, Chen J, Zhou YB, Chen M, Ma YZ, Xu ZS, Cui XY (2021) Genome-wide analysis of the DUF4228 family in soybean and functional identification of GmDUF4228 -70 in response to drought and salt stresses. *Front Plant Sci* 12:628299–628299. <https://doi.org/10.3389/fpls.2021.6282991>
- Li J, Hu E, Chen X, Xu J, Lan H, Li C, Hu Y, Lu Y (2016) Evolution of DUF1313 family members across plant species and their association with maize photoperiod sensitivity. *Genom* 107:199–207. <https://doi.org/10.1016/j.ygeno.2016.01.0031>
- Li L, Xie C, Ye T, Xu J, Chen R, Gao X, Zhu J, Xu Z (2017) Molecular characterization, expression pattern and function analysis of the rice OsDUF866 family. *Biotechnol Biotechnol Equip* 31:243–249. <https://doi.org/10.1080/13102818.2016.12689321>
- Li LH, Lv MM, Li X, Ye TZ, He X, Rong SH, Dong YL, Guan Y, Gao XL, Zhu JQ, Xu ZJ (2018) The rice OsDUF810 family: OsDUF810.7 may be involved in the tolerance to salt and drought. *Mol Biol* 52:489–496. <https://doi.org/10.1134/S002689331804012X1>
- Li L, Ye T, Guan Y, Lv M, Xie C, Xu J, Gao X, Zhu J, Cai L, Xu Z (2018) Genome-wide identification and analyses of the rice OsDUF936 family. *Biotechnol Biotechnol Equip* 32:309–315. <https://doi.org/10.1080/13102818.2017.14134211>
- Liu C, Xie T, Chen C, Luan A, Long J, Li C, Ding Y, He Y (2017) Genome-wide organization and expression profiling of the R2R3-MYB transcription factor family in pineapple (*Ananas comosus*). *BMC Genom* 18:1–16. <https://doi.org/10.1186/s12864-017-3896-y1>
- Maher C, Stein L, Ware D (2006) Evolution of arabidopsis microRNA families through duplication events. *Genome Res* 16:510–519. <https://doi.org/10.1101/gr.46805061>
- Marciniak K, Przedniczek K (2019) Comprehensive insight into gibberellin- and jasmonate-mediated stamen development. *Genes* 10:811. <https://doi.org/10.3390/genes101008111>
- Montalbini P (1995) Effect of rust infection on purine catabolism enzyme levels in wheat leaves. *Physiol Mol Plant Pathol* 46:275–292. <https://doi.org/10.1006/pmpp.1995.10221>
- Nabi RBS, Tayade R, Imran QM, Hussain A, Shahid M, Yun B-W (2021) Functional insight of nitric-oxide induced DUF genes in arabidopsis. *Nitric Oxide Plants*. <https://doi.org/10.3389/fpls.2020.010411>

- Nietzsche M, Schiebl I, Börnke F (2014) The complex becomes more complex: protein-protein interactions of SnRK1 with DUF581 family proteins provide a framework for cell- and stimulus type-specific SnRK1 signaling in plants. *Front Plant Sci*. <https://doi.org/10.3389/fpls.2014.000541>
- Niron H, Barlas N, Salih B, Türet M (2020) Comparative transcriptome, metabolome, and ionome analysis of two contrasting common bean genotypes in saline conditions. *Front Plant Sci*. <https://doi.org/10.3389/fpls.2020.5995011>
- Olvera Carrillo Y, Campos F, Reyes JL, Garcarrubio A, Covarrubias AA (2010) Functional analysis of the group 4 late embryogenesis abundant proteins reveals their relevance in the adaptive response during water deficit in arabidopsis. *Plant Physiol* 154:373–390. <https://doi.org/10.1104/pp.110.1589641>
- Qiao X, Li Q, Yin H, Qi K, Li L, Wang R, Zhang S, Paterson AH (2019) Gene duplication and evolution in recurring polyploidization–diploidization cycles in plants. *Genome Biol* 20:1–23. <https://doi.org/10.1186/s13059-019-1650-21>
- Qiu L, Chen R, Fan Y, Huang X, Luo H, Xiong F, Liu J, Zhang R, Lei J, Zhou H, Wu J, Li Y (2019) Integrated mRNA and small RNA sequencing reveals microRNA regulatory network associated with internode elongation in sugarcane (*Saccharum officinarum* L.). *BMC Genom* 20:817. <https://doi.org/10.1186/s12864-019-6201-41>
- Rai A, Suprasanna P, D'Souza SF, Kumar V (2012) Membrane topology and predicted RNA-binding function of the 'early responsive to dehydration (ERD4)' plant protein. *PLoS ONE*. <https://doi.org/10.1371/journal.pone.00326581>
- Sagi M, Omarov RT, Lips SH (1998) The Mo-hydroxylases xanthine dehydrogenase and aldehyde oxidase in ryegrass as affected by nitrogen and salinity. *Plant Sci* 135:125–135. [https://doi.org/10.1016/S0168-9452\(98\)00075-21](https://doi.org/10.1016/S0168-9452(98)00075-21)
- Shaik R, Ramakrishna W (2012) Bioinformatic analysis of epigenetic and microRNA mediated regulation of drought responsive genes in rice. *PLoS ONE* 7:e49331. <https://doi.org/10.1371/journal.pone.00493311>
- Shao H, Chen S, Zhang K, Cao Q, Zhou H, Ma Q, He B, Yuan X, Wang Y, Chen Y (2014) Isolation and expression studies of the ERD15 gene involved in drought-stressed responses. *Genet Mol Res* 13:10852–10862. <https://doi.org/10.4238/2014.December.19.61>
- Vaattovaara A, Brandt B, Rajaraman S, Safronov O, Veidenberg A, Luklová M, Kangasjärvi J, Löytynoja A, Hothorn M, Salojärvi J, Wrzaczek M (2019) Mechanistic insights into the evolution of DUF26-containing proteins in land plants. *Commun Biol*. <https://doi.org/10.1038/s42003-019-0306-91>
- Wang L, Shen R, Chen LT, Liu YG (2014) Characterization of a novel DUF1618 gene family in rice. *J Integr Plant Biol* 56:151–158. <https://doi.org/10.1111/jipb.121301>
- Wang T, Chen L, Zhao M, Tian Q, Zhang W-H (2011) Identification of drought-responsive microRNAs in *Medicago truncatula* by genome-wide high-throughput sequencing. *BMC Genom* 12:367. <https://doi.org/10.1186/1471-2164-12-3671>
- Wang Y, Tang H, DeBarry JD, Tan X, Li J, Wang X, Lee T-h, Jin H, Marler B, Guo H (2012) MCSanX: a toolkit for detection and evolutionary analysis of gene synteny and collinearity. *Nucleic Acids Res* 40:e49–e49. <https://doi.org/10.1093/nar/gkr12931>
- Waseem M, Aslam MM, Shaheen I (2021) The DUF221 domain-containing (DDP) genes identification and expression analysis in tomato under abiotic and phytohormone stress. *GM Crop Food*. <https://doi.org/10.1080/21645698.2021.19622071>
- Watanabe S, Matsumoto M, Hakomori Y, Takagi H, Shimada H, Sakamoto A (2014) The purine metabolite allantoin enhances abiotic stress tolerance through synergistic activation of abscisic acid metabolism. *Plant Cell Environ* 37:1022–1036. <https://doi.org/10.1111/pce.122181>
- Yang Q, Niu X, Tian X, Zhang X, Cong J, Wang R, Zhang G, Li G (2020) Comprehensive genomic analysis of the DUF4228 gene family in land plants and expression profiling of ATDUF4228 under abiotic stresses. *BMC Genom* 21:12. <https://doi.org/10.1186/s12864-019-6389-31>
- Yuan F, Yang H, Xue Y, Kong D, Ye R, Li C, Zhang J, Theprungsirikul L, Shrift T, Krichilsky B (2015) OSCA1 mediates osmotic-stress-evoked Ca²⁺ increases vital for osmosensing in *Arabidopsis*. *Nature*, vol 514, pg 367
- Zhong H, Zhang H, Guo R, Wang Q, Huang X, Liao J, Li Y, Huang Y, Wang Z (2019) Characterization and functional divergence of a novel DUF668 gene family in rice based on comprehensive expression patterns. *Gene* 10:980. <https://doi.org/10.3390/genes101209801>
- Zhou X, Zhu X, Shao W, Song J, Jiang W, He Y, Yin J, Ma D, Qiao Y (2020) Genome-wide mining of wheat DUF966 gene family provides new insights into salt stress responses. *Front Plant Sci*. <https://doi.org/10.3389/fpls.2020.5698381>

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